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GLIBCXX_3.4.20 not found

Supported OS

- Linux x64
- MacOS x64
- Windows x64

Installation

Linux

Excute setup script `cle-xxx.linux.bin`

```
chmod +x ./cle-xxx.linux.bin
./cle-xxx.linux.bin
```

Start a new terminal or run the following command

```
source ~/.bash_profile # for Bash
# or maybe `source ~/.bashrc`

# source ~/.zshrc # for Zsh
```

Uninstall

```
cle uninstall
```

Macos

Excute setup script `cle-xxx.macos.bin`

```
chmod +x ./cle-xxx.macos.bin
./cle-xxx.macos.bin
```

Start a new terminal or run the following command

```
source ~/.bash_profile # for Bash
# or maybe `source ~/.bashrc`

# source ~/.zshrc # for Zsh
```

Uninstall

```
cle uninstall
```

Windows

Dependencies

- [Microsoft Visual C++ Redistributable for Visual Studio 2019](#)

Run `cle-xxx.win64.exe`

Configurations

Copy the following template to a file named `.cle.json5` in your home directory.

Path for Unix-like OS: `~/.cle.json5`

Path for Windows: `%userprofile%\cle.json5`

```
{
  // Logs directory.
  // $HOME redirects to the current user's home directory.
  // $TMP redirects to the operating system's default directory for
  temporary files
  LOGS_PATH: '$TMP/cle/logs',

  // When a file size becomes higher than this value it will rotate it
  (its possible that
  // the worker check the file after it actually pass the limit) . You
  can specify the
  // unit at then end: 10G, 10M, 10K
  LOG_MAX_SIZE: '1G',

  // Enable compression via gzip for all rotated logs
  LOG_COMPRESS: true,

  // This cron is used to a force rotate when executed. We are using
  node-schedule to
  // schedule cron, so all valid cron for node-schedule is valid cron
  for this option.
  LOG_ROTATE_INTERVAL: '0 0 * * *',
```

```
// Duration per file for saving buffers. (ms)
BUFFERS_DURATION_PER_FILE: 600000,

// Duration per file for saving records. (ms)
RECORDS_DURATION_PER_FILE: 600000,

// Network Interface
// eg. 'eth0', 'wlan'
NETWORK_INTERFACES: ['eth0'],

// HTTP API Port
HTTP_API_PORT: 44444,

// UDP API Port
UDP_API_PORT: 55555,

// Beacon Lifetime
BEACON_LIFETIME: 21600000,

// Max number of UDP Subscribers
MAX_NUMBER_OF_UDP_SUBSCRIBERS: 1,
}
```

Usage

Note: Under Windows, you must run the command as an administrator.

Activate CLE

```
cle activate path/to/license
```

Eg.

```
cle activate ~/Downloads/cle.license
```

Service commands

Start

```
cle service start [path/to/cpaFile] [flags]
```

Eg.

```
cle service start ~/Document/demo.cpa
```

Stop

```
cle service stop
```

Logs

Realtime

```
cle service logs
```

Logs save paths

Logs path can be modified by editing `config.json5` .

```
// Logs directory.  
// $HOME redirects to the current user's home directory.  
// $TMP redirects to the operating system's default directory for  
temporary files  
LOGS_PATH: '$TMP/cle/logs',
```

Flush logs

```
cle service flush
```

Status

```
cle service status
```

More

```
cle help
```

HTTP APIs

Project Info

GET <http://localhost:44444/info>

HTTP Status

Code	Description	Body
200	OK	JSON Result

Result Structure

```
{
  id: string; // Project id
  name: string; // Project name
  isPusher: boolean; // Use pusher
  managerApiUrl: string; // RTLS Server URL
  managerKey: string; // RTLS Server key
  managerSecret: string; // RTLS Server secret
  category: null | string; // Project category
  version: string; // CPA version
  maps: string[]; // Maps
  algorithms: string[]; // Algorithms
}
```

Example. GET <http://localhost:44444/info>

```
{
  "id": "HAX51aCkA",
  "name": "test",
  "isPusher": true,
  "managerApiUrl": "http://api-ltt.rtls.dev.corelink.vip",
  "managerKey": "teohoSpnVt",
  "managerSecret": "Rt7262333913",
  "category": null,
  "version": "0.2.0",
  "maps": [
    "duIKAGcQ0w"
  ],
  "algorithms": [
    "AOA",
    "RSSI"
  ]
}
```


Map

GET <http://localhost:44444/info/maps/:mapId>

HTTP Status

Code	Description	Body
200	OK	JSON Result
404	Not Found	Error Message

Result Structure

```
{
  id: string; // map ID
  name: string; // map name
  areas: Array<{ // areas info
    id: string; // area ID
    name: string; // area name
    points: Array<[number, number, number, number]> // area coordinates.
    [x, y, z, open].
    lines: Array<{ // routes info
      id: string; // route ID
      name: string; // route name
      points: Array<[number, number, number]> // route coordinates [x,
y, z]
      connections: string[]; // connections
    }>;
    connections: string[]; // connections
    useRoutes: boolean; // use routes
  }>;
  gateways: Array<A0AGateway | RSSIGateway>; // gateways info
  width: number; // map width in meters
```

```

height: number; // map height in meters
useGps: boolean; // use GPS
gpsOriginLat?: number; // GPS latitude
gpsOriginLon?: number; // GPS longitude
gpsOffsetDegrees?: number; // offset degree
gpsOriginMap?: [number, number]; // GPS coordinate in map
originPixelX: number; // origin x in picture
originPixelY: number; // origin y in picture
widthInPixels: number; // map width in pixels
heightInPixels: number; // map height in pixels
metersPerPixel: number; // meters per pixel
model: { // model
  type: 'image' | 'collada' | 'zip'; // type
  size: string; // file size
  extra: { // Extras. eg. textures.
    path: string; // path
    size: string; // file size
  };
};
};
}

```

Example. GET <http://localhost:44444/info/maps/duIKAGcQOw>

```

{
  "id": "duIKAGcQOw",
  "name": "test1",
  "areas": [
    {
      "id": "adNGXd8mv",
      "name": "办公区",
      "points": [
        [
          0.07,
          5.444,

```



```
"width": 50,  
"height": 50,  
"useGps": true,  
"gpsOriginLat": 50,  
"gpsOriginLon": 50,  
"gpsOffsetDegrees": 0,  
"gpsOriginMap": [  
  0,  
  0  
],  
"originPixelX": -412,  
"originPixelY": 512,  
"widthInPixels": 1024,  
"heightInPixels": 1024,  
"metersPerPixel": 0.048828125,  
"model": {  
  "type": "image",  
  "base64": "..."  
},  
}
```

Model

GET <http://localhost:44444/info/maps/:mapId/model>

HTTP Status

Code	Description	Body
200	OK	Binary
404	Not Found	Error Message

Model Base64

GET <http://localhost:44444/info/maps/:mapId/model/base64>

HTTP Status

Code	Description	Body
200	OK	JSON Result
404	Not Found	Error Message

Result Structure

```
{
  base64: string;
}
```

Algorithms

GET <http://localhost:44444/info/algorithms>

HTTP Status

Code	Description	Body
200	OK	JSON Result

Result Structure

```
[AOA, RSSI]
```

Example. GET <http://localhost:44444/info/algorithms>


```
"hmmBeta": 2,
"hmmSigma": 0.5,
"staticDataSwitch": false,
"staticDataNum": 1000,
"allStaticData": false,
"timeLog": false,
"generateJson": false,
"file": "data/123",
"beacons": [
  {
    "id": "cxInUq_1qM",
    "mac": "",
    "pLog": false,
    "coordinates": [
      0,
      0
    ]
  }
]
```

List all beacons

GET <http://localhost:44444/beacons>

Parameters

Name	In	Optional	Description
macs	query	YES	Beacon macs

HTTP Status

Code	Description	Body
200	OK	JSON Result

Result Structure

```
{
  [beaconMac: string]: {
    x: number; // Position x-axis
    y: number; // Position y-axis
    z: number; // Position z-axis
    rssi: number; // RSSI value
    mapId: string; // Map ID
    zoneId: string; // Zone ID
    gps: null | { // WGS-84 coordinates
      lat: number; // latitude
      lon: number; // longitude
    };
    updatedAt: number; // Timestamp for last package
    calculatedAt: number; // Timestamp after calculation
    userData: {
      0: {
        channel: number; //
        frequency: number; //
        recv: boolean;
        power: { //
          type: 'TI' | 'Nordic'; //
          value: number; //
        };
        sos: boolean; // Button pressed
        battery: number; // Battery percentage
        lastPressedTS: number; // Timestamp button pressed
      };
    };
    userDataTs: {
```

```

    [cmd: number]: number; // userData timestamp
  };
  lastGateway: string; // Gateway mac
}
}

```

Example. GET <http://localhost:44444/beacons?macs=806fb00c45d9,806fb00c45c9>

```

{
  "806fb00c45d9": {
    "x": 9.573189735412598,
    "y": 5.213486194610596,
    "z": 0.800000011920929,
    "rssi": -60.00,
    "mapId": "lwlx6fFgu",
    "zoneId": "z1",
    "gps": null,
    "updatedAt": 1571109529308,
    "calculatedAt": 1571109529309,
    "userData": {
      "0": {
        "channel": 2481,
        "frequency": 50,
        "power": {
          "type": "TI",
          "value": 5
        },
      },
      "sos": false,
      "battery": 83
    }
  },
  "userDataTs": {
    "0": 1571109529201
  }
}

```

```
    },
    "lastGateway": "80:6f:b0:c4:13:60"
  },
  "806fb00c45c9": {
    "x": 6.329273700714111,
    "y": 7.880806922912598,
    "z": 0.800000011920929,
    "rssi": -60.00,
    "mapId": "lwLx6fFgu",
    "zoneId": "z1",
    "gps": null,
    "updatedAt": 1571109529215,
    "calculatedAt": 1571109529216,
    "userData": {
      "0": {
        "channel": 2481,
        "frequency": 110,
        "power": {
          "type": "TI",
          "value": 0
        },
      },
      "sos": false,
      "battery": 95
    }
  },
  "userDataTs": {
    "0": 1571109529201
  },
  "lastGateway": "80:6f:b0:c4:13:60"
}
```

List all locators

GET <http://localhost:44444/locators>

Parameters

Name	In	Optional	Description
macs	query	YES	Locator macs

HTTP Status

Code	Description	Body
200	OK	JSON Result

Result Structure

```
{
  [locatorIpAddressV4: string]: {
    mac: string; // Locator mac
    angles: number[6]; // Locator angles
    updatedAt: number; // Timestamp for last package
    online: boolean; // online
    info?: {
      name: string; // Locator name
      type: string; // Locator type
      mapId: string; // Map Id
      zoneId: string; // Zone Id
      channel: string; // Working channel
      elevation: number; // Elevation for AOA algorithm
      azimuth: number; // Azimuth for AOA algorithm
      rotation: number; // Rotation for AOA algorithm
      errorDegree: number; // Error Degree for AOA algorithm
    }
  }
}
```

```
x: number; // Position x-axis
y: number; // Position y-axis
z: number; // Position z-axis
};
}
```

Example. GET <http://localhost:44444/locators?macs=3c:fa:d3:b0:33:f8,3c:fa:d3:b0:33:fc>

```
{
  "192.168.124.60": {
    "mac": "3c:fa:d3:b0:33:f8",
    "angles": [
      553,
      392,
      -1840,
      427,
      -268,
      990
    ],
    "updatedAt": 1582795155178,
    "info": {
      "x": 12,
      "y": 2,
      "z": 3,
      "mapId": "63dhjtywP",
      "elevation": 2.766,
      "azimuth": 219.278,
      "rotation": 325.054,
      "errorDegree": 0.357,
      "channel": null,
      "name": "1",
      "type": "AOA",
    }
  }
}
```


Retrieve a beacon

GET <http://localhost:44444/beacons/:mac>

Parameters

Name	In	Optional	Description
mac	path	NO	Beacon mac

HTTP Status

Code	Description	Body
200	OK	JSON Result
404	Not Found	Error Message

Result Structure

```
{
  mac: string; // Beacon mac
  x: number; // Position x-axis
  y: number; // Position y-axis
  z: number; // Position z-axis
  rssi: number; // RSSI value
  mapId: string; // Map ID
  zoneId: string; // Zone ID
  gps: null | { // WGS-84 coordinates
    lat: number; // latitude
    lon: number; // longitude
  };
  updatedAt: number; // Timestamp for last package
  calculatedAt: number; // Timestamp after calculation
}
```



```

userData: {
  0: {
    channel: number; //
    frequency: number; //
    power: { //
      type: 'TI' | 'Nordic'; //
      value: number; //
    };
    sos: boolean; // Button pressed
    battery: number; // Battery percentage
  };
};
userDataTs: {
  [cmd: number]: number; // userData timestamp
};
lastGateway: string; // Gateway mac
}

```

Example. GET <http://localhost:44444/beacons/806fb00c45c9>

```

{
  "mac": "806fb00c45c9",
  "x": 6.963463306427002,
  "y": 6.864238739013672,
  "z": 0.800000011920929,
  "rssi": -60.00,
  "mapId": "lwlx6fFgu",
  "zoneId": "z1",
  "gps": null,
  "updatedAt": 1571120302882,
  "calculatedAt": 1571120302883,
  "userData": {
    "0": {
      "channel": 2481,

```

```
    "frequency": 110,
    "power": {
      "type": "TI",
      "value": 0
    },
    "sos": false,
    "battery": 95
  },
  "userDataTs": {
    "0": 1571109529201
  },
  "lastGateway": "80:6f:b0:c4:13:60"
}
```

Example. GET http://localhost:44444/beacons/mac_not_existed

Beacon not found

Retrieve a locator

GET <http://localhost:44444/locators/:mac>

Parameters

Name	In	Optional	Description
mac	path	NO	Locator mac

HTTP Status

Code	Description	Body
200	OK	JSON Result
404	Not Found	Error Message

Result Structure

```
{
  ip: string; // Locator ip address v4
  mac: string; // Locator mac
  angles: number[6]; // Locator angles
  updatedAt: number; // Timestamp for last package
  online: boolean; // online
  info?: {
    name: string; // Locator name
    type: string; // Locator type
    mapId: string; // Map Id
    zoneId: string; // Zone Id
    channel: string; // Working channel
    elevation: number; // Elevation for AOA algorithm
    azimuth: number; // Azimuth for AOA algorithm
    rotation: number; // Rotation for AOA algorithm
    errorDegree: number; // Error Degree for AOA algorithm
    x: number; // Position x-axis
    y: number; // Position y-axiy
    z: number; // Position z-axis
  };
}
```

Example. GET <http://localhost:44444/locators/3c:fa:d3:b0:33:f8>

```
{
  "ip": "192.168.124.60",
```

```
"mac": "3c:fa:d3:b0:33:f8",
"angles": [
  553,
  392,
  -1840,
  427,
  -268,
  990
],
"updatedAt": 1582795155178,
"info": {
  "x": 12,
  "y": 2,
  "z": 3,
  "mapId": "63dhjtywP",
  "elevation": 2.766,
  "azimuth": 219.278,
  "rotation": 325.054,
  "errorDegree": 0.357,
  "channel": null,
  "name": "1",
  "type": "AOA",
  "zoneId": null
}
}
```

Example. GET http://localhost:44444/locators/mac_not_existed

Locator not found

Send beacon event

POST <http://localhost:44444/beacons/:mac/event>

Parameters

Name	In	Optional	Description
mac	path	NO	Beacon mac
json	body	NO	JSON Body

HTTP Status

Code	Description	Body
200	OK	Empty
400	Failed	Empty
404	Not Found	Error Message

JSON Body Structure

```
{
  value: number; // Event value
  duration?: number; // Event duration (s)
  ext?: number[3]; // Extension
}
```

Example. POST <http://localhost:44444/beacons/ada29b164768/event>

```
{
  "type": "beaconEvent",
  "value": 1,
  "duration": 20
}
```

Custom Userdata Cache

GET <http://localhost:44444/beacons/:mac/custom/:cmd>

Parameters

Name	In	Optional	Description
mac	path	NO	Beacon mac
cmd	path	NO	Custom command

HTTP Status

Code	Description	Body
200	OK	JSON Result

JSON Body Structure

```
Array<[cmd, value1, value2, value3, timestamp]>
```

Example. GET <http://localhost:44444/beacons/3cfad3b09998/custom/9>

```
[[9,17,34,21,1584686032298],[9,17,34,24,1584686032335],
[9,17,34,27,1584686032377],[9,17,34,30,1584686032426],
[9,17,34,33,1584686032474],[9,17,34,39,1584686032554],
```

[9,17,34,42,1584686032602],[9,17,34,45,1584686032663],
[9,17,34,54,1584686032782],[9,17,34,57,1584686032828],
[9,17,34,60,1584686032860],[9,17,34,66,1584686032958],
[9,17,34,69,1584686033006],[9,17,34,78,1584686033140],
[9,17,34,81,1584686033182],[9,17,34,87,1584686033264],
[9,17,34,93,1584686033361],[9,17,34,96,1584686033401],
[9,17,34,2,1584686033489],[9,17,34,11,1584686033620],
[9,17,34,14,1584686033667],[9,17,34,17,1584686033714],
[9,17,34,20,1584686033752],[9,17,34,23,1584686033803],
[9,17,34,29,1584686033892],[9,17,34,32,1584686033931],
[9,17,34,35,1584686033981],[9,17,34,38,1584686034021],
[9,17,34,41,1584686034072],[9,17,34,44,1584686034111],
[9,17,34,47,1584686034151],[9,17,34,50,1584686034201],
[9,17,34,56,1584686034286],[9,17,34,59,1584686034332],
[9,17,34,62,1584686034382],[9,17,34,68,1584686034465],
[9,17,34,74,1584686034556],[9,17,34,80,1584686034641],
[9,17,34,86,1584686034738],[9,17,34,89,1584686034780],
[9,17,34,92,1584686034812],[9,17,34,95,1584686034866],
[9,17,34,98,1584686034909],[9,17,34,1,1584686034957],
[9,17,34,7,1584686035041],[9,17,34,10,1584686035085],
[9,17,34,13,1584686035141],[9,17,34,22,1584686035264],
[9,17,34,25,1584686035311],[9,17,34,28,1584686035345],
[9,17,34,37,1584686035491],[9,17,34,40,1584686035533],
[9,17,34,43,1584686035572],[9,17,34,46,1584686035620],
[9,17,34,55,1584686035760],[9,17,34,61,1584686035845],
[9,17,34,64,1584686035880],[9,17,34,70,1584686035976],
[9,17,34,76,1584686036059],[9,17,34,79,1584686036111],
[9,17,34,82,1584686036160],[9,17,34,91,1584686036286],
[9,17,34,0,1584686036412],[9,17,34,12,1584686036600],
[9,17,34,18,1584686036687],[9,17,34,21,1584686036738],
[9,17,34,24,1584686036768],[9,17,34,27,1584686036828],
[9,17,34,33,1584686036916],[9,17,34,39,1584686036995],
[9,17,34,42,1584686037045],[9,17,34,45,1584686037092],
[9,17,34,48,1584686037132],[9,17,34,54,1584686037221],
[9,17,34,57,1584686037271],[9,17,34,60,1584686037301],
[9,17,34,66,1584686037402],[9,17,34,69,1584686037450],

[9,17,34,72,1584686037480],[9,17,34,75,1584686037528],
[9,17,34,81,1584686037624],[9,17,34,84,1584686037661],
[9,17,34,87,1584686037711],[9,17,34,90,1584686037760],
[9,17,34,93,1584686037801],[9,17,34,96,1584686037838],
[9,17,34,99,1584686037882],[9,17,34,2,1584686037939],
[9,17,34,8,1584686038013],[9,17,34,11,1584686038067],
[9,17,34,14,1584686038113],[9,17,34,17,1584686038174],
[9,17,34,20,1584686038189],[9,17,34,23,1584686038239],
[9,17,34,26,1584686038302],[9,17,34,32,1584686038404],
[9,17,34,35,1584686038417],[9,17,34,38,1584686038465],
[9,17,34,44,1584686038544],[9,17,34,47,1584686038595],
[9,17,34,50,1584686038645],[9,17,34,53,1584686038689],
[9,17,34,59,1584686038772],[9,17,34,62,1584686038818],
[9,17,34,65,1584686038866],[9,17,34,71,1584686038947],
[9,17,34,74,1584686039027],[9,17,34,80,1584686039083],
[9,17,34,83,1584686039128],[9,17,34,86,1584686039176],
[9,17,34,95,1584686039302],[9,17,34,1,1584686039417],
[9,17,34,4,1584686039450],[9,17,34,13,1584686039580],
[9,17,34,16,1584686039608],[9,17,34,19,1584686039656],
[9,17,34,22,1584686039711],[9,17,34,25,1584686039761],
[9,17,34,28,1584686039787],[9,17,34,34,1584686039882],
[9,17,34,37,1584686039931],[9,17,34,43,1584686040012],
[9,17,34,46,1584686040060],[9,17,34,49,1584686040115],
[9,17,34,55,1584686040201],[9,17,34,58,1584686040240],
[9,17,34,61,1584686040288],[9,17,34,64,1584686040323],
[9,17,34,70,1584686040418],[9,17,34,73,1584686040518],
[9,17,34,79,1584686040558],[9,17,34,88,1584686040678],
[9,17,34,91,1584686040728],[9,17,34,94,1584686040780],
[9,17,34,97,1584686040838],[9,17,34,0,1584686040877],
[9,17,34,3,1584686040910],[9,17,34,6,1584686040959],
[9,17,34,9,1584686041004],[9,17,34,15,1584686041083],
[9,17,34,18,1584686041133],[9,17,34,24,1584686041212],
[9,17,34,27,1584686041281],[9,17,34,30,1584686041317],
[9,17,34,33,1584686041359],[9,17,34,36,1584686041404],
[9,17,34,39,1584686041440],[9,17,34,42,1584686041488],
[9,17,34,45,1584686041534],[9,17,34,48,1584686041570],


```
[9,17,34,51,1584686041616],[9,17,34,54,1584686041671],  
[9,17,34,57,1584686041714],[9,17,34,60,1584686041745],  
[9,17,34,63,1584686041798],[9,17,34,69,1584686041890],  
[9,17,34,72,1584686041935],[9,17,34,75,1584686041972],  
[9,17,34,78,1584686042035],[9,17,34,81,1584686042068],  
[9,17,34,84,1584686042109],[9,17,34,87,1584686042150],  
[9,17,34,90,1584686042218],[9,17,34,93,1584686042249],  
[9,17,34,96,1584686042282]]
```

Remove outdated beacons

GET <http://localhost:44444/flush-beacons>

HTTP Status

Code	Description	Body
200	OK	JSON Result

Result Structure

```
{  
  count: number; // number of beacons removed  
}
```

Example. GET <http://localhost:44444/flush-beacons>

```
{  
  "count": 10  
}
```

UDP Subscription

Subscribe

nodejs code example

```
import { createSocket } from 'dgram';

const client = createSocket('udp4');
client.send('subscribe', 55555, '127.0.0.1');
client.on('message', msg => {
  try {
    const message = JSON.parse(msg.toString());
    console.log(message);
  } catch (e) {
    console.error(e);
  }
});
```

Unsubscribe

nodejs code example

```
import { createSocket } from 'dgram';

const client = createSocket('udp4');
client.send('unsubscribe', 55555, '127.0.0.1');
```

Message

Structure

```
{
  type: 'sensors' | 'locators';
  data: SensorData | LocatorData;
}
```

Example 1: Sensors

```
{
  "type": "sensors",
  "data": {
    "806fb00c45d9": {
      "x": 9.573189735412598,
      "y": 5.213486194610596,
      "z": 0.800000011920929,
      "mapId": "lw1x6fFgu",
      "zoneId": "z1",
      "gps": null,
      "updatedAt": 1571109529308,
      "calculatedAt": 1571109529309,
      "userData": {
        "0": {
          "channel": 2481,
          "frequency": 50,
          "power": {
            "type": "TI",
            "value": 5
          },
          "sos": false,
          "battery": 83
        }
      },
      "userDataTs": {
        "0": 1571109529201
      }
    }
  }
}
```

```

    "lastGateway": "80:6f:b0:c4:13:60"
  },
  "806fb00c45c9":{
    "x":6.329273700714111,
    "y":7.880806922912598,
    "z":0.800000011920929,
    "mapId":"lwlx6fFgu",
    "zoneId":"z1",
    "gps": null,
    "updatedAt":1571109529215,
    "calculatedAt": 1571109529216,
    "userData":{
      "0":{
        "channel":2481,
        "frequency":110,
        "power":{
          "type":"TI",
          "value":0
        },
        "sos":false,
        "battery":95
      }
    },
    "userDataTs": {
      "0": 1571109529201
    },
    "lastGateway": "80:6f:b0:c4:13:60"
  }
}

```

Example 2: Locators

```
{
```



```

        "x": 2,
        "y": 1,
        "z": 1,
        "mapId": "63dhjtywP",
        "elevation": 175.796,
        "azimuth": 302.003,
        "rotation": 278.412,
        "errorDegree": 0.197,
        "channel": null,
        "name": "6",
        "type": "AOA",
        "zoneId": null
    }
}
}
}

```

SensorData

```

{
  [beaconMac: string]: {
    x: number; // Position x-axis
    y: number; // Position y-axis
    z: number; // Position z-axis
    mapId: string; // Map ID
    zoneId: string; // Zone ID
    gps: null | { // WGS-84 coordinates
      lat: number; // latitude
      lon: number; // longitude
    };
    updatedAt: number; // Timestamp for last package
    calculatedAt: number; // Timestamp after calculation
    userData: {
      0: {
        channel: number; //

```

```

    frequency: number; //
    recv: boolean;
    power: { //
      type: 'TI' | 'Nordic'; //
      value: number; //
    };
    sos: boolean; // Button pressed
    battery: number; // Battery percentage
    lastPressedTS: number; // Timestamp button pressed
  };
};
userDataTs: {
  [cmd: number]: number; // userData timestamp
};
lastGateway: string; // Gateway mac
}
}

```

LocatorData

```

{
  [locatorIpAddressV4: string]: {
    mac: string; // Locator mac
    angles: number[]; // Locator angles
    updatedAt: number; // Timestamp for last package
    online: boolean; // online
    info?: {
      name: string; // Locator name
      type: string; // Locator type
      mapId: string; // Map Id
      zoneId: string; // Zone Id
      channel: string; // Working channel
      elevation: number; // Elevation for AOA algorithm
      azimuth: number; // Azimuth for AOA algorithm
      rotation: number; // Rotation for AOA algorithm
    }
  }
}

```

```
    errorDegree: number; // Error Degree for AOA algorithm
    x: number; // Position x-axis
    y: number; // Position y-axiy
    z: number; // Position z-axis
  };
};
}
```

Beacon event

nodejs code example

```
import { createSocket } from 'dgram';

const client = createSocket('udp4');
client.send(JSON.stringify({
  type: 'beaconEvent',
  value: 1,
  duration: 5,
  mac: 'ada29b164768',
}), 55555, '127.0.0.1');
```

Debug

Debug logs

```
cle service debug onloff
```

Eg.

```
cle service debug on
```

```
cle service debug off
```

Buffers

```
cle service save buffers onloff [path/to/dir]
```

Eg.

```
cle service save buffers on
```

```
cle service save buffers off
```

Records

```
cle service save records onloff [path/to/dir]
```

Eg.

```
cle service save records on
```

```
cle service save records off
```

Startup Script Generator (Unix-like OS)

CLE uses PM2 to generate startup scripts and configure them in order to keep your process list intact across expected or unexpected machine restarts.

Init systems supported

- ☒ systemd: Ubuntu >= 16, CentOS >= 7, Arch, Debian >= 7
- ☐ upstart: Ubuntu <= 14
- ☒ launchd: Darwin, MacOSx
- ☐ openrc: Gentoo Linux, Arch Linux
- ☐ rcd: FreeBSD
- ☐ systemv: Centos 6, Amazon Linux
- ☒ registry: Windows 10

Generating a startup script

To get the automatically-configured startup script for your machine you need to type this command:

```
cle pm2 startup
```

Do not pass sudo to this command! It will print the exact right command you will have to copy/paste into your terminal

Example:

```
$ cle pm2 startup
[PM2] Init System found: systemd
[PM2] To setup the Startup Script, copy/paste the following command:
sudo env PATH=$PATH:/home/you/.corelink/cle cle pm2 startup systemd -u you
--hp /home/you
```

You simply have to copy/paste the line PM2 gives you and the startup script will be configured for your OS.

NOTE 1 : When upgrading to newer CLE version, update the PM2 startup script! Use `cle pm2 unstartup` first then `cle pm2 startup` again

NOTE 2: You can customize the service name via the `--service-name` option

Saving current process list

Important

Once you started all the applications you want to manage, you have to save the list you wanna respawn at machine reboot with:

```
cle pm2 save
```

It will save the process list with the corresponding environments into the dump file
`~/.corelink/cle/pm2/dump.pm2`

Disabling startup system

```
cle pm2 unstartup
```

FAQ

GLIBCXX_3.4.20 not found

For low version Ubuntu

```
sudo add-apt-repository ppa:ubuntu-toolchain-r/test  
sudo apt-get update  
sudo apt-get install libstdc++6
```